

Team Master Builders

Yo-Yo Project Addendum Report

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2.008: Design & Manufacturing II

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0. Acknowledgements

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We would also like to thank Professor Kait Becker and Dr. John Liu, and the TAs Daniel Massimino, Seiji Engelkemier, and Minna Wyttenbach for teaching us the foundations of manufacturing and equipping us with the knowledge and fortitude to excel in the world of manufacturing.

1. Design Overview and Team Organization

Team B's yo-yo design was inspired by LEGO's iconic yellow minifigure head. After scaling the dimensions of the existing product, the two halves of the yo-yo were designed to represent the top and bottom portions of the face. The original concept included an interactive feature, which was a cutout in the top injection-molded body of the yo-yo where the eyes would be. This opening served as a window to the thermoformed part underneath, which was intended to be rotated by the user to change the minifigure's eye expression. However, due to challenges with achieving a reliable press-fit and the increasing complexity of the required geometry, this feature was removed during redesign.

Eliminating the cutout simplified the yo-yo to three unique injection-molded parts with identical top and bottom bodies. The top ring was designed to house the thermoformed part, while the bottom ring consisted of supporting walls for the yo-yo's structure. The thermoformed part is inspired by the LEGO stud commonly seen on LEGO bricks, with the team members' initials replacing the logo. The final design retained the classic LEGO head's expression, with the facial features spray-painted onto the bodies using vinyl sticker stencils. All injection-molded and thermoformed parts were color-matched to the characteristic LEGO minifigure yellow to ensure a cohesive appearance in the final yo-yo.



Figure 1: Final Yo-Yo Picture. Credit: Professor Josh and his exquisite photography skills.

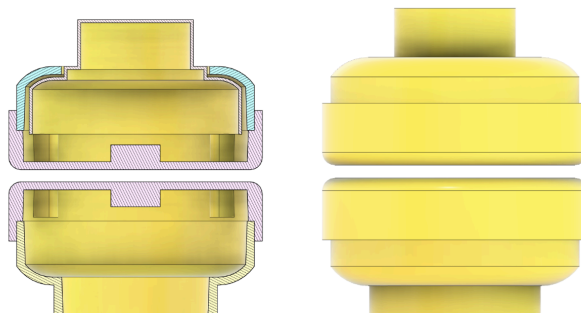


Figure 2: Section analysis view of the final yo-yo architecture.

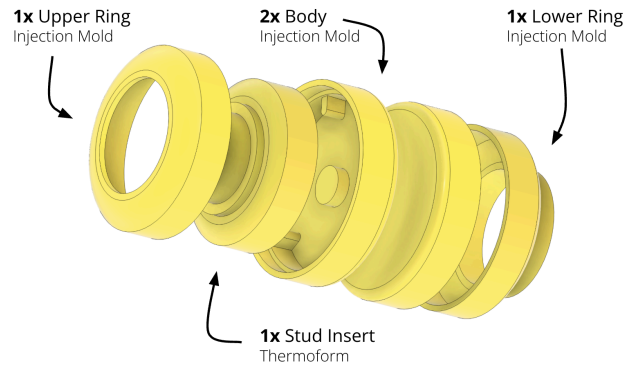


Figure 3: Exploded view of the final yo-yo assembly.

Table 1: Bill of Materials (BOM) for an individual yo-yo.

Part	Description	Source	Unit Cost	Quantity	Total Cost
Body	Injection-molded polypropylene (PP); press fits with rings, one holds thermoformed piece	LMP	N/A	2	N/A
Top Ring	Injection molded polypropylene (PP); press fit with body on the top of the yo-yo	LMP	N/A	1	N/A
Bottom Ring	Injection molded polypropylene (PP); press fit with body on the bottom of the yo-yo	LMP	N/A	1	N/A
Thermoform Piece	Yellow nylon and polyethylene terephthalate glycol (PETG)	LMP	N/A	1	N/A
Yellow Vinyl Wrap	Applied to thermoform part; cost allocated per yo-yo	Amazon	\$0.53	1	\$0.53
Sticker Stencil	Used as a mask to spray-paint facial features; not included in final assembly	N52	N/A	1	N/A
Black Spray Paint	Used to paint facial features; cost allocated per yo-yo	Amazon	\$0.1238	1	\$0.1238
10-24, 5/8" set screw	Holds two bodies together	LMP (McMaster)	\$0.0741	1	\$0.0741
10-24 Nut	Insert molded into body	LMP (McMaster)	\$0.0404	2	\$0.0808
Yo-Yo String	Wrapped around center of yo-yo	LMP	\$0.1993	1	\$0.1993
Total					\$1.508

1.1 Team Organization

Throughout the semester, tasks were divided between team members at each stage to maintain a consistent workload. After each lab meeting, we created a to-do list and distributed responsibilities so work could be completed in parallel. During the design stage, the injection-molded parts and the thermoformed part were split among team members (Angel, Avishai, Lanie, and Huda) to complete the CAD and CAM, while Teagan managed the manufacturing spreadsheet and parameterization. Thermal analysis of the injection-molded parts was completed by Angel and Teagan to prepare for manufacturing and to motivate any final design adjustments.

For the design review, when the first injection mold pair was due, Angel machined the bottom injection-molded body, while the other members prepared the presentation. Subsequent manufacturing processes were completed based on availability, with all team members rotating through shifts during the reserved slots for CNC milling, injection molding, and thermoforming. Mass production and assembly followed a similar structure, with reservations planned at the beginning of each week during lab. Measurement and tolerance analysis were completed by Angel and Avishai. For both the poster and the report, the drafting and editing responsibilities for specific sections were divided among the team to ensure timely completion.

2. Manufacturing and Parameter Analysis

2.1 Manufacturing Spreadsheet Results and Parameter Edits

Several sections of the yo-yo were altered throughout the process. The initial yo-yo design was not able to be manufactured because of the small geometries, so the design was changed to have the ring press fit inside the body. In an effort to reduce the height of the thermoform part and decrease complexity, the part was placed on top of ribs inside the body instead of being sandwiched between the ring and body. The size of the ribs was determined by possible tool paths. Initially, the press fit interference was estimated to be 0.015 inches by replicating prior yo-yo interferences. The expectation was that the initial fit would not be sufficient, and the part could be remachined to increase interference. Instead, the press fit was too tight, and the molds needed to be remachined. Due to a dwindling supply of mold blanks, the new approach was to have no tolerance between the parts and slowly machine the mold larger until a desired fit was achieved. After injection molding the parts with no tolerance to check the fit, the parts fit together sufficiently, and therefore, the interference was set to zero. The top ring had some consistency issues where it would pop out of the body, so 0.002 additional inches were machined off its diameter to improve fit.

Shrinkage of the part is most important at the press fit area. The press fit shrinkage is perpendicular to the flow of plastic. Using the figure from the 2.008 learning sequences, lateral

polypropylene shrinkage is 2%. Volumetric shrinkage simulations confirmed this, with shrinkage around the main portion of the press fit generally consistent with the predicted value (Figure 5). Therefore, the shrinkage compensation decided on for each part was 2%. The simulation also predicted less shrinkage around the edges of the part, which was later reflected when the zero-interference parts fit together. The reduced shrinkage around the edge allowed for a tighter-than-expected press fit.

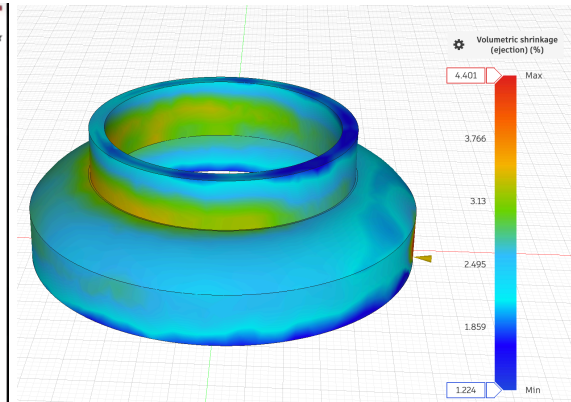
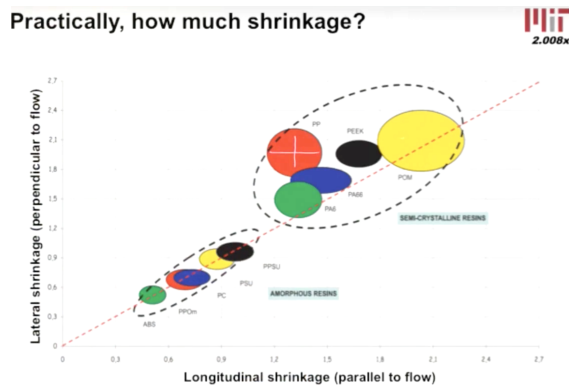


Figure 4: Material shrinkage chart from video lectures.

Figure 5: Shrinkage simulation analysis conducted through Fusion.

CMM values of the press fit section after machining were 0.013% larger than the top ring CAD, 0.023% smaller than the bottom ring CAD, and 0.112% larger than the body CAD. The larger variation between expected and actual values for the body is due to a rough finishing machining pass, which created more variance on the surface of the mold.

Table 2: Values at the press fit interference for each injection molded part.

	Initial Design Specifications	Shrinkage Compensation for Mold (2%)	CMM Measured Molds
Top Ring	2.398 ± 0.005 in	2.447 ± 0.005 in	2.44732 ± 0.00001
Body	2.340 ± 0.005 in	2.449 ± 0.005 in	2.44973 ± 0.00001
Bottom Ring	2.398 ± 0.005 in	2.447 ± 0.005 in	2.44844 ± 0.00001

2.2 Injection Molding Process Parameters

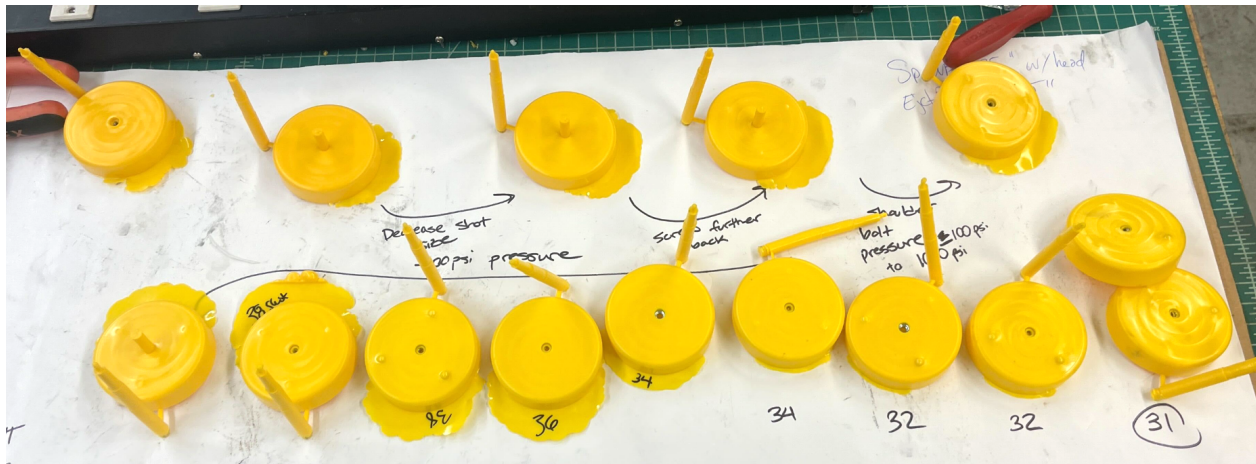


Figure 6: Progression of part quality while varying injection molding parameters

During the injection molding process, the Boy was the primary injection molding machine used. Initially, the sprue hole into the mold wasn't filed out, and there was insufficient packing pressure. Because of this, plastic did not fully inject into the mold, resulting in several short shots. In addition, cooling time was fine-tuned to 15 seconds for the body and 12 seconds for the ring, so the plastic was solid enough to eject but not too difficult to remove due to shrinkage. After packing pressure was increased to 480 psi for the body and 500 psi for the rings, the dosage stroke for each part was fine-tuned to prevent flashing. For the body, this was 40 mm, and for the rings 30 mm. This process was repeated with the inclusion of overmolded nuts in the body, reducing mold size and dosage stroke to 31mm (Figure 6). When initially determining the quantity of colorant, the plastic was glassy, so the colored pellet quantity was increased to 5% to guarantee consistent opacity. The pellet mix that best matched the color of the thermoformed part was a ratio of 12.8 g of 21, 14.3 g of 20, and 3 g of 13 to about 600 g of clear pellets.

3. The Journey

3.1 Troubleshooting: The Expression Window

Having extensively analyzed the press-fit geometry and dimensions, we now turn to a less-prominent yet nearly as challenging aspect of our yo-yo's design: the expression window. In the spirit of LEGO, our vision was to integrate a "play feature" into the yo-yo, which took the form of exchangeable facial expressions. As mentioned previously, we were forced to forgo the idea due to time constraints; this section will analyze the troubleshooting process leading up to that point.

We began with a rectangular 0.3" by 1.2" window that would display multiple eyes and eyebrows for the minifigure. However, this led to a discontinuity in both the ring and the body press-fit and caused deep and narrow cavities in the cavity mold. We overcame the former issue by shifting up the window to only adjust the eyebrow expression, which made the body press-fit continuous and as a result improved the rigidity of this side of the press-fit. We resolved the latter issue by inverting and simplifying the press-fit geometry, enabling us to move the parting line and eliminate the difficult-to-machine pockets. We also added lateral protrusions to the core mold (along with matching cutouts in the cavity mold) to prevent plastic from leaking into the window.

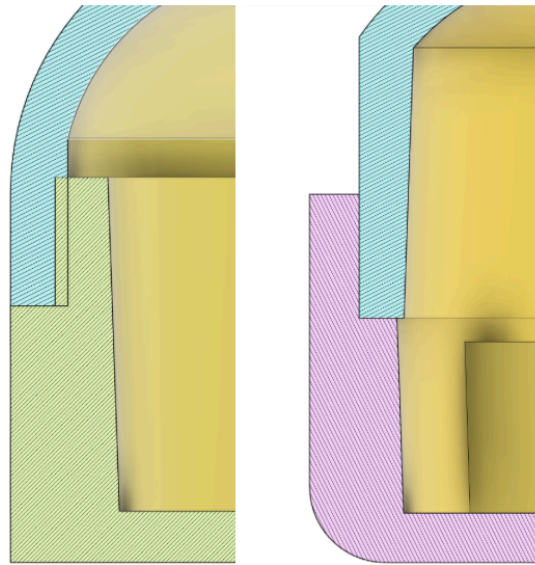


Figure 7: Comparison of press-fit geometries, with the initial version on the left and the final version on the right. Note the inverted arrangement and the more consistent wall thickness on the final design.

The team pressed on and successfully injection-molded with the updated molds. Yet despite this DFM and manufacturing-related troubleshooting, we ran into new issues with buckling and a generally unreliable press fit as a result of the compliant nature of the thermal plastic. Two potential solutions were to: remove the window and make both press-fit faces continuous, or to stiffen the wall with ribs. The latter approach interfered with the thermoformed part, and with little time to ideate further we pursued the former option, resulting in a successful and reliable press-fit.

3.2 Design Decisions and Project Management

Our team converged on a vision during the first lab, and due to it referencing a well-known design, it was easy to communicate and work on. We verbally defined our intended scope, complexity, and backup plan, but did not formalize beyond that. At times, this lack of formality slowed us down when deciding whether to scale back the project scope, but even so, we successfully scaled down and redesigned twice.

In retrospect, the most valuable suggestion we would make to our past selves would be to implement a project planning strategy to organize the team's operations. Our team would have greatly benefited from a Gantt chart, as this would have helped us keep track of both immediate

and longer-term goals and deadlines. An alternative suggestion is to designate a team leader who could supply this organization to the team, though we favor the Gantt chart approach due to the potential difficulty in establishing trust in a leader within the short project timespan.

3.3 Design for Assembly (DFA)

The majority of our team's efforts were focused on DFM and a successful press-fit. However, this also led to a simplified design compared to our initial idea and made it easier to assemble. This section will focus on the next steps our team would take to intentionally apply DFA to reduce assembly time and complexity if given more time.

The most problematic part of the assembly process was cutting the thermoformed parts down to size—it is slow and introduces significant variation. The key lesson here is to reduce reliance on manual operations in manufacturing, especially when it comes to critical dimensions.

The thermoformed part forms the stud on the minifigure head, and requires rather accurate height trimming. Too tall, and it prevents proper seating of the press-fit. Too short, and the part rattles within the injection-molded shell. This is further exacerbated by the fact that the rings were manufactured across multiple runs and different ejector pins were selected, causing variation in the ejector pin ridges; this means the optimal height of the thermoformed part depends on the ring it is paired with. To resolve this, we propose the following two changes: the addition of a flange at the base of the thermoformed part and the adoption of a standard ejector pin size.

Adding a flange to the lower edge of the thermoformed part would mean that the height of the thermoformed part is determined by the mold geometry and not our manual cuts. This is far more consistent and significantly speeds up the cutting operation, as this cut wouldn't be creating a critical dimension, instead merely trimming excess material.

A standard ejection pin allows the use of the aforementioned thermoformed part with pre-determined height. However, this still relies on good structured communication within the team. We would go further by measuring the thickness of the mudframe and available ejector pins to set one size nearly flush with the mold surface, essentially eliminating the bump caused by the ejector pins. This would make an optimal pin size and reduce ambiguity in the pin length choice. We believe that in tandem, these two measures would effectively let us reach a consistent part geometry and an easy-to-assemble yo-yo.

4. Cost Analysis and Production Scaling

4.1 Yo-Yo Materials Cost Analysis

At our current manufacturing scale of roughly 50 yo-yos, concerns with cost have been minimal. Labor cost is non-existent as our employees are compensated with points and positive feedback, overhead cost is likewise not a consideration, and neither is cost of central materials. To scale up production to the millions, however, it is necessary to account for the many avenues where cost can tank a business. Our yo-yos consist of three unique injection molded parts, one of which has an overmolded nut, and a thermoformed insert. Each of these components is associated with a set of material, and manufacturing costs

Each yo-yo required 3.113 in^3 of a 5% pigment and polypropylene mixture. Polypropylene costs \$0.19/ lb [4-1] at a density of 866 kg/m^3 [4-3].

$$C_{\text{polypropylene}} = \frac{\text{volume}}{\text{yo-yo}} * \frac{\text{mass}}{\text{volume}} * \frac{\text{cost}}{\text{mass}}$$

$$C_{\text{polypropylene}} = \frac{5.101 \text{ e-}5 \text{ m}^3}{\text{yo-yo}} * \frac{866 \text{ kg}}{\text{m}^3} * \frac{\$0.419}{\text{kg}} = \$0.0185$$

Each yo-yo also has two overmolded nuts and a set screw to hold the two sides together which cost \$0.0404/nut and \$0.0741/set screw, and the string for the yo-yo which costs \$.1993/string.

$$C_{\text{hardware}} = 2 * C_{\text{nut}} + C_{\text{screw}} + C_{\text{string}} = \$0.3542$$

Additionally, the 5" x 5" x 0.030" PETG thermoplastic sheets used cost roughly \$0.019/in² so roughly \$0.48/sheet [4-2].

$$C_{\text{thermoplastic}} = \$0.48$$

To match the thermo formed portion to the rest of the yo-yo, dual cast wrap vinyl was applied before the thermo forming process. At \$0.026/in² and a 4.5" x 4.5" sheet necessary for each part, it cost roughly \$0.53 per yo-yo.

$$C_{\text{vinyl}} = \$0.53$$

The cost due to spray painting will not be included in this analysis, as for higher volume production a higher fidelity and efficiency process would be preferred. Therefore the total material costs are:

$$C_{\text{polypropylene}} + C_{\text{thermoplastic}} + C_{\text{vinyl}} + C_{\text{hardware}} = \$1.3842$$

Additional contributing costs are labor costs, and overhead costs due to machining. The 6 members of our team spent roughly 4 hours a week beyond the required lab hours for 11 weeks designing and manufacturing 56 yo-yos. Minimum wage in Massachusetts is \$15 so labor cost per yo-yo at our scale would be:

$$C_{labor} = \frac{\$15}{hr} * \frac{18 + 4 hrs}{week} * \frac{11 weeks}{56 yo-yos} = \$64.8$$

These don't account for the cost of the team consultation with the pappalardo shop staff or the cost of maintenance and part replacement. Our team alone broke at least two carbide lathe tools, a probe, and an end mill, all of which in a stand alone factory setting would contribute to overall expected costs.

4.2 Scaled Production Plan

To scale our production system to withstand the manufacturing requirements of a million yo-yos, we need to rethink our assembly pipeline. A flow shop style factory would be ideal for high volume yo-yo production, where not much flexibility is required. Four manufacturing stages, and a central assembly station would create an efficient system for yo-yo production. The four stages would produce injection molded bodies, top rings, bottom rings, and thermoformed tops in parallel. Each station has its own associate buffer, and post production processing stations, where the sprues and any additional flash can be removed by workers from the injection molded parts, and the thermoformed parts can be cut with a die with a machine. Additionally, buffer space can be controlled by altering the number of cavities per mold. For example, the consumption of bodies is double that of each of the rings since it is a duplicate part. To account for this, the body core and cavity may be designed to accommodate two parts per round of injection molding.

In terms of equipment, we would need to purchase enough machines per station to output one million yo-yos when running in parallel. Based on our prior yo-yo assembly experience, we found the longest process was producing the thermoformed piece. Assuming in our factory, with automation, it takes 30 seconds to produce a thermoformed part, and the machine works for 10 hours a day with 90% uptime, then the number of parts one thermoform machine can produce in a year is:

One thermoformed machine can produce:

$$S_{thermoformed} = \frac{9 hours}{1 day} * \frac{3600 seconds}{1 hour} * \frac{1 part}{30 seconds} * \frac{365 days}{1 year} = 394,200 parts per year$$

If the thermoform machine can only produce 400,000 parts per year, then we need three thermoformed machines in order to reach one million parts per year.

For the injection molding machines, assuming same parameters but only taking 25 seconds per part, we would need three IM machines per station to reach one million parts:

$$S_{IM} = \frac{9 hours}{1 day} * \frac{3600 seconds}{1 hour} * \frac{1 part}{25 seconds} * \frac{365 days}{1 year} = 473,040 parts per year$$

The additional costs for the machines adds cost per yo-yo. Assuming we buy production-grade injection molding machines at \$50,000 per machine [4-7] and thermoform machines for \$40,000 [4-8], rough average of prices found online, then the costs comes out to be

$$C_{equipment} = \frac{1}{473040 \text{ ppy (w/ uptime cost)}} * \frac{\$50,000}{20 \text{ years}} * 3 \text{ machines} * 3 \text{ parts}$$

$$\frac{1}{394200 \text{ ppy}} * \frac{\$40,000}{20 \text{ years}} * 3 \text{ machines} = \$0.06$$

The main tooling cost comes from the injection molding machine molds. The molds in the lab currently are aluminum. This was chosen because aluminum is relatively cheap and easy to machine. However, aluminum possesses a relatively high thermal expansion constant and relatively low hardness. This means that in a high volume production setting, the yo-yos coming out of the aluminum mold when the machine is first turned on will vary greatly with the yo-yos coming out even a few hours later. The solution would be to switch to a steel alloy for our mold material. This would decrease our variation, which we struggled with in the lab, as shown in section 4.3. We would need to have 12 sets of machine molds for each station, 4 per machine in each station in order to reach production of one million yo-yos, assuming each mold can last 100,000 cycles [4-5]. For the thermoform mold, we would have to purchase 12 aluminum thermoformed molds, assuming each mold lasts a cycle of 100,000 thermoformed pieces. This would be a switch from the SLA-printed molds we have in the lab, which comes from the ability for aluminum molds to last much longer than SLA-printed molds in a production environment. We would purchase the molds and pay an upfront cost for the molds to be made through a third-party vendor. It would not be cost-efficient to purchase an industrial CNC mill when we only need to make one type of mold. We are assuming a typical set of steel molds costs \$20,000 and an aluminum thermoforming mold costs \$2,000 [4-5, 4-6]. Assuming a set of tools consists of one mold for each stage, the tooling cost for each yo-yo equals:

$$C_{tooling} = \frac{\$20,000}{1000000 \text{ parts}} * 12 + \frac{\$2000}{1000000 \text{ parts}} * 12 = \$0.26$$

At the million product scale the expected final material cost converges to the material cost per yo-yo found in section 4.1, which we found to be \$1.38/yo-yo. However to minimize the material cost we could switch to pre-colored high impact polystyrene sheets. This would not only reduce the labor cost associated with adhering the cast vinyl, but it would also reduce the material and machining cost associated with repeated use of a higher cutting force.

Because not everything is automated we would have to include a labor cost for production. We would have 2 employees per station to survey the production and troubleshoot any issues. The reason for having workers as well is that we found during lab that sometimes there may be issues a machine may not understand, like if the injection molded part is not ejected properly, which could cause a lot of damage to the machine and slow down production rate. By having 2 employees per station, we would be able to accurately assess any issues that arise and insure the highest quality parts.

Putting together all these costs, our estimated price per yo-yo, not including labor costs and overhead, is:

$$C_{yo-yo} = C_{tooling} + C_{material} + C_{equipment} = \$1.70 \text{ per yoyo}$$

4.3 Statistical Analysis for Interference Probability

Knowing that we had issues within our low volume production to create the correct interference fit, we'd imagine that we may run into similar issues when running a high-volume production plan. We therefore want to provide the probability that if a ring and body were to be chosen at random, that they would fall within specification for the interference fit. These probabilities would inform us in regards to how many more yo-yos we need to produce in order to statistically produce one million yo-yos that are fit for customer use.

After measuring the outer diameter of 50 parts of both the bottom and top ring, as well as the inner diameter of the body, we found each parts respective mean and standard deviation to produce the following measurements:

$$OD_{top\ ring} = 2.3994 \pm 0.0133\ inches$$

$$OD_{bottom\ ring} = 2.4000 \pm 0.0073\ inches$$

$$ID_{body} = 2.4004 \pm 0.0068\ inches$$

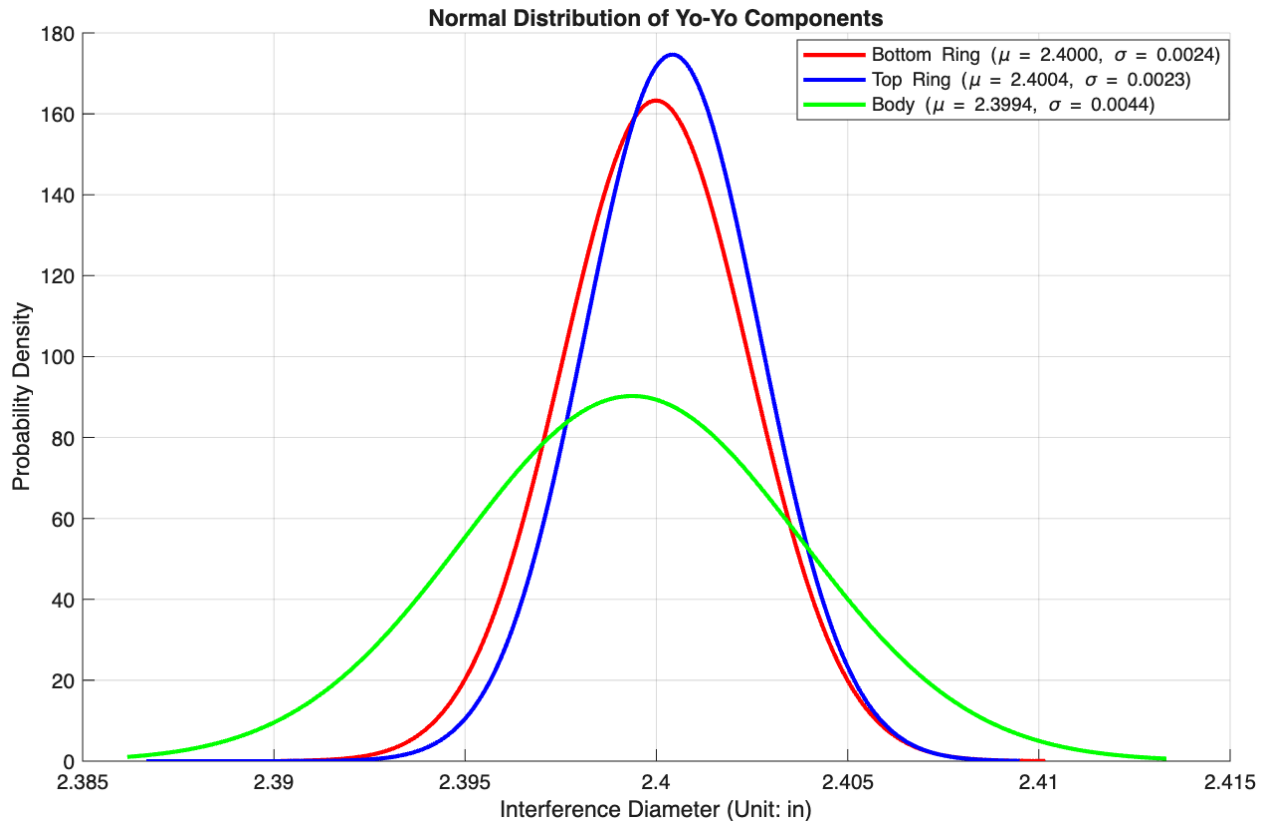


Figure 8: Normal distribution curves of measurements of yo-yo components as measured in Lab.

4.3.1 Bottom Ring to Body Interference Fit Probability

$$\mu_{fit} = \mu_{ring} - \mu_{body} = 2.400 - 2.3994 = 0.0006\ inches$$

$$\sigma_{fit} = \sqrt{\sigma_{ring}^2 + \sigma_{body}^2} = \sqrt{.00244^2 + .00442^2} = .00505 \text{ inches}$$

$$z_{upper} = \frac{.002 - .0006}{.00505} = .27$$

$$z_{lower} = \frac{0 - .0006}{.00505} = -.12$$

$$\text{Interference Probability} = \Phi(z_{upper}) - \Phi(z_{lower}) = 60.8\% - 45.1\% = 15.7\%$$

4.3.2 Top Ring to Body Interference Fit Probability

$$\mu_{fit} = \mu_{ring} - \mu_{body} = 2.4004 - 2.3994 = 0.00105$$

$$\sigma_{fit} = \sqrt{\sigma_{ring}^2 + \sigma_{body}^2} = \sqrt{.00228^2 + .00442^2} = .00498$$

$$z_{upper} = \frac{.002 - .00105}{.00498} = .19$$

$$z_{lower} = \frac{0 - .00105}{.00498} = -.21$$

$$\text{Interference Probability} = \Phi(z_{upper}) - \Phi(z_{lower}) = 57.53\% - 41.60\% = 15.93\%$$

4.3.3 Probability Implications

Lowering our variance and increasing tolerance fit probability are key parts of our goal to produce one million yo-yos. As discussed earlier, we would change the material of our molds to a material that can withstand the environment of high volume production. Additionally, we would spend more time working on the CAM of our molds to ensure we have accurately machined molds. We would also think of increasing the interference fit to make sure more parts fit. With these changes in mind, we think we can successfully scale to production of one million yo-yos.

Citations:

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